Electric car: launch the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Plan a safe team workflow and identify the car's energy pathway.

SUCCESS CRITERIA - I can...

1. Label the battery, switch, motor and connections.
2. Identify safety checkpoints before tools or power.
3. Connect a design choice with a measurable outcome.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "prototype" and "constraint" to explain your answer. Sentence starters are optional.

prototype: a version built to test a design

constraint: a requirement or limit that the design must meet

evidence: observations or measurements that support an idea

01 Warm-up: bring back an earlier idea

A single-loop toy calculation gives **3 V** induced emf. **Explain why this alone cannot** predict the effects on a whole power grid; **name two missing features**.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: average speed = distance/elapsed time. Use your own design and measurements. Battery → motor → drivetrain. Follow teacher training for tools and hot materials.

Worked reasoning: You can test a statement about a design if it names something you can measure. "The car is fast" can become "The car travels **3 m** in under **2 s** on this level track." This requires an average speed above **1.5 m/s**.

Your turn: Draw a test track with start, finish, distance and timing method labeled.

Sentence starter: In my work, I used … to show … . I labeled it … .

Start Finish



Distance: ____ Timing method: _____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Sketch your proposed car**.
2. **Label the battery**, switch, motor, driven axle and wheels.
3. **Add a theme** that will not obstruct moving parts.

Sentence starter: In my work, I used … to show … . I labeled it … .



Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

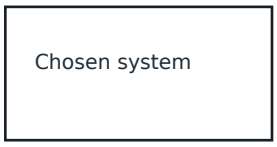
04 Apply the physics

Your first design uses a **3 V** supply and an expected motor current of **0.40 A**.
1. **Estimate electrical input power**.
2. Label at least three destinations that prevent all input power from becoming useful forward-motion power.

Word help: estimate = find an approximate value

Sentence starter: I predict … . I used … to make this prediction because … .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

1. **Write one measurable design goal** and one requirement or limit.
2. **Explain how you** will measure the goal instead of judging only by appearance.

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A circuit has **4** identical series resistors. **Add two more** at unchanged ideal supply voltage. **Compare new current** with the original.
Draw the stated circuit and **label quantities relevant** to the question.

Word help: current = rate of charge flow

Sentence starter: Both … . They differ in … because … .

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Create a fair first-test plan naming distance, timing method, number of trials and variables held fixed.

Sentence starter: My design includes … . It does what is needed by … .

Check / criterion	Measurement	Decision rule
___	___	___
___	___	___

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Choose a speed target** and a reliability target for a **3 m** course.
2. Design repeated trials and a rule for judging success without selecting only the fastest run.

Sentence starter: My design includes … . It does what is needed by … .

Check / criterion	Measurement	Decision rule
___	___	___
___	___	___

Marzano check - circle: 1 2 3